EXPERIMENT 14 — Pro-Code Extension Using Zoho Catalyst

Aim

To study how pro-code development can extend low-code applications using Zoho Catalyst and understand its role in implementing advanced functionalities in the EnergyWise application.

Theory

Pro-code development involves writing conventional programming code to implement complex functionality that may not be conveniently supported by visual development tools alone.

Zoho Creator provides low-code features such as forms, reports and workflows. Zoho Catalyst provides additional development capabilities for building custom application components and services.

The combination of low-code and pro-code development can be useful when an application requires advanced processing, external integrations or specialized algorithms.

For the EnergyWise application, pro-code extensions could support:

1. Energy prediction: Forecasting future energy consumption using historical readings.
2. Advanced analytics: Analysing consumption patterns and identifying unusual energy usage.
3. External integration: Connecting the application to building sensors or third-party energy management systems.
4. Custom processing: Performing calculations or data transformations beyond the built-in workflow requirements.
5. Scalability: Separating specialized processing tasks into independently managed services where appropriate.

Proposed architecture

Zoho Creator

Forms, reports and workflows

API communication

Transfers authorized data between services

Zoho Catalyst / Custom code

Specialized processing and integration

Energy analytics service

Optional forecasting and prediction model

Note: This is a proposed architecture for the project, not an implemented integration.

Procedure

1. Study the concept of pro-code development and its differences from low-code development.
2. Explore the purpose and capabilities of Zoho Catalyst.
3. Identify advanced EnergyWise requirements that could benefit from custom code.
4. Design an architecture connecting Zoho Creator with an external processing service through an API.
5. Study how the external service could process energy data and return results to the application.
6. Document the advantages and limitations of this approach.

Observation

Aspect	Low-code approach	Pro-code extension

Application forms	Visual form builder	Custom interfaces if needed
Integration	Supported connectors and APIs	Custom API clients and services

Result

The role of pro-code development in extending low-code applications was studied. A proposed architecture combining Zoho Creator, API communication and Zoho Catalyst was designed to demonstrate how advanced processing and machine-learning capabilities could be incorporated into the EnergyWise application.